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RESEARCH ARTICLE

Multi-Scene Dataset and Object Detector for Outside Blind Individual Identification

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ABSTRACT In the field of Computer Vision (CV), assistive navigation systems for visually impaired individuals have garnered significant attention in recent years. However, most existing solutions rely on wearable devices or mobile platforms, which often face limitations in cost, deployment, and robustness in complex outdoor environments. This paper proposes a practical approach to public space recognition for the visually impaired. The proposed recognition method can be widely applied to existing public video surveillance systems. We created a specialized image dataset tailored for recognition by visually impaired individuals. At the same time, for recognition tasks involving nighttime environments and partially occluded targets, we propose a composite framework that integrates various grouped convolutions and image enhancement networks. Compared to the original baseline detection models, our models outperform on our created dataset by 1.5% average precision (e.g., from 94.1% to 95.6% for YOLOv8x), while reducing parameters by up to 35.7% (e.g., from 56.9M to 36.6M for YOLOv11x). Furthermore, our models also achieve over 0.8% AP and a parameter reduction exceeding 10% compared to the original baseline models on ExDARK dataset.

INDEX TERMS Machine learning, object detection, visually impaired, white cane.

I. INTRODUCTION

With the global population of visually impaired individuals continuously increasing, research on assistive technologies for this group has gained significant momentum. A considerable amount of recent work focuses on leveraging computer vision techniques and portable mobile devices to provide assistive functionalities for the blind [1]. Over the past decade, substantial efforts have been made to develop visual aid systems for blind individuals. In particular, the widespread adoption of artificial intelligence algorithms in recent years has offered more reliable technological support for navigation systems [2]. However, the market still lacks portable, highly accurate, and cost-effective outdoor solutions that are accessible to the majority of visually impaired individuals. This situation can be attributed to challenges in software algorithms, hardware technologies such as specialized chips, and concerns regarding cost, safety, and reliability [3].

In our previous work [4], we analyzed the challenges faced by blind individuals in outdoor navigation and the unique characteristics of their outdoor activities. We proposed a novel approach that utilizes smart city public surveillance systems to identify blind individuals and provide them with assistance through public infrastructure. This approach has relatively low technical implementation complexity and, in theory, can be realized simply by analyzing and recognizing images captured by surveillance systems. It is applicable to nearly all outdoor public scenarios. To facilitate this functionality, we created an outdoor scenes dataset tailored to the characteristics of visually impaired individuals. We validated the reliability of this dataset using several mainstream object detection systems. However, our prior work also revealed significant limitations in existing datasets and detection models, such as low recognition accuracy in low light environments.

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In recent years, numerous fusion-based improvements have been proposed for YOLO [5], [6], [7]. YOLO-based outdoor recognition systems—such as pedestrian detection and license plate recognition—have also been extensively studied and widely applied [8], [9], [10], [11]. Despite significant progress in object detection, most state-of-the-art detectors still struggle in low-light and partially occluded environments. Studies on datasets like ExDARK [12] and Occluded-DukeMTMC [13] have shown considerable drops in detection accuracy compared to well-lit settings, indicating the need for models specifically designed to tackle these conditions. For detection tasks under low-illumination conditions, numerous image enhancement networks achieving state-of-the-art performance have been proposed in recent years [14], [15], [16], [17]. An introduction to these methods is provided in Sections II-B and IV-B1, along with a comparative analysis of their performance based on experimental results. Note that not all of these networks are compatible well with YOLO-based systems, and many of the enhancement modules introduce significant additional computational overhead. For improving occluded object detection in YOLO-based systems, current baseline models primarily focus on optimizing the loss function and the data augmentation algorithms [18], [19]. Nevertheless, detecting partially occluded objects remains one of the most significant challenges faced by modern object detection systems.

This paper proposes a novel fusion-based improvement framework that is compatible with all major YOLO baseline systems. We introduce the Context Guide Block (CGBlock) [20] Bottleneck module and the Spatially Enhanced Attention Module (SEAM) [21] based on various grouped convolutions to enhance the detector's robustness in recognizing targets under low illumination and partial occlusion. Additionally, we integrate a low-light enhancement network and a newly designed feature fusion network, which not only improves recognition performance in low-light environments but also contributes to overall model lightweighting.

In addition to optimizing the object detector, we constructed a novel image dataset specifically for outdoor feature detection aimed at assisting visually impaired individuals. This dataset includes over 500 outdoor scenes and contains numerous scenarios simulating the conditions of visually impaired individuals walking in low-light environments. The dataset comprehensively covers challenging scenarios that can be encountered in outdoor feature recognition tasks for visually impaired individuals.

We evaluated the reliability of the proposed optimized models. Experimental results indicate that mainstream object recognition models can achieve satisfactory performance when trained on our proposed dataset. In comparison to baseline methods, our proposed model delivers more precise and computationally efficient recognition of features related to visually impaired individuals in complex outdoor environments. We also validated the generalizability of our proposed optimization framework on a widely used open-source low-light dataset. Our approach not only consistently outperforms the baseline models in low-light object recognition tasks, achieving higher accuracy with lower computational cost, but also contributes to overall model lightweighting.

Key contributions of this study are summarized as follows:

- A unique dataset specifically tailored for blind person recognition in diverse outdoor public scenarios has been constructed. This dataset emphasizes the identification of features associated with blind canes and includes a variety of challenging images depicting blind individuals in different activities and environments.
- We propose an optimization framework for mainstream convolutional neural network systems, designed for object detection tasks in outdoor environments with occlusions and low-light conditions. The framework incorporates a Context Guided block-inspired structure within the backbone network, integrates the Retinexformer network and the Spatially Enhanced Attention Module (SEAM), and further employs a novel Cross-Scale Feature Fusion network to enhance multi-scale representation.
- Our models achieve notable improvements in detection accuracy while simultaneously reducing model parameter count and computational overhead. Compared to the baseline models, our models achieve an improvement of at least 0.7% AP on both our created dataset and the ExDARK dataset, while reducing the parameter count by at least 14.3% and computational requirements by at least 6.8%.

The remaining parts of this article are organized as follows. Section II presents the related work, including key challenges in outdoor blind individual recognition and an overview of current object detection methods under challenging scenarios. Section III introduces the dataset we constructed, along with another publicly available dataset used in this study. Section IV provides a detailed explanation of the proposed optimization framework. Section V describes the experimental setup and workflow. Section VI reports and analyzes the results of our extensive experiments. Section VII concludes the study with a summary of findings and outlines directions for future research. Section VIII provides the method for obtaining our dataset and the license for the source code. Section IX demonstrates the strict protocols we adhere to in ethical matters.

II. RELATED WORK

A. CHALLENGES OF OUTDOOR BLIND FEATURE RECOGNITION

A significant challenge in identifying blind individuals in public spaces lies in effective and identifiable features. Our previous study [4] revealed that physiological traits, such as the eyes or stride length of blind individuals, are not suitable as primary identification targets for object detection systems. Consequently, we selected the white cane, a uniquely characteristic item commonly carried by blind individuals, as the primary identification class. In addition, we observed that outdoor environments include objects resembling white canes, such as walking sticks used by elderly individuals and long-handled umbrellas. To address this, our dataset incorporates these items as interference categories to enhance robustness.

Identifying blind individuals in outdoor environments shares similarities with pedestrian detection research [22]. Key issues also arise in low-light conditions, crowded public spaces where the target is partially occluded, and environments with complex artificial lighting. Our previous study [4] indicated that these factors significantly reduce detection accuracy. Moreover, the distinctive design of white canes presents additional challenges. The warning sections of white canes are often bright red, a color that can appear distorted under artificial lighting at night due to color shifts caused by complex illumination conditions.

To address these challenges, we expanded our dataset to include a variety of new images. These include scenarios where individuals holding white canes are captured in low-light environments, in crowded public spaces where the cane is partially occluded, and in nighttime settings with complex artificial light sources. These additions significantly enhance the diversity and applicability of our updated dataset. Meanwhile, the optimization framework proposed in the following sections enhances the recognition system's performance in these more challenging scenarios.

B. SPECIAL SCENE OBJECT DETECTION

In recent years, enhancement networks designed to improve object detection performance under challenging conditions—such as low-light, foggy, and occluded environments—have received increasing attention alongside the rapid development of object recognition systems. Representative approaches include the Self-Calibrated Illumination (SCI) Learning Framework [14], which utilizes a multi-level Retinex structure with shared weights and a self-calibration module to restore structural details; the Illumination Adaptive Transformer (IAT) network [15], which simultaneously utilizes multi-layer convolution and Transformer attention layers to reconstruct local and global image illumination information; the Retinexformer network [16], which incorporates Retinex theory withSelf-attention mechanism to achieve effective low-light image enhancement; and the Gated Differentiable Image Processing (GDIP) network [17], which integrates differentiable image processing operations with weighted combinations to selectively enhance features relevant to detection. We conducted an analysis of several state-of-the-art (SOTA) frameworks published in the past two years, comparing their applicability to different scenarios, performance, and computational complexity. Based on this analysis, we aimed to select the most suitable enhancement framework as the data augmentation network for the input stage (Section IV-B1).

To improve the detection of occluded objects, most mainstream baseline object detectors primarily focus on

TABLE 1. Statistics of our new dataset.

Class	Number of Images	Number of Types	Number of Public Scenes
Blind stick	1502	2	294
Normal stick	362	126	109
Umbrella	151	69	52

TABLE 2. Statistics of the dataset created in our previous study [4].

Class	Number of Images	Number of Types	Number of Public Scenes
Blind stick	780	2	149
Normal stick	110	60	51
Umbrella	110	50	38

enhancements at the image input stage and the detection head. For instance, input-level data augmentation techniques such as Cutout and Mosaic were introduced as early as YOLOv4 and YOLOv5 [18], [19]. In recent years, increasingly complex loss functions like CIoU [23], SIoU [24], and WIoU [25] have also been adopted in detection head designs. In contrast, we suggest that the incorporating attention mechanisms can strengthen the model's ability to recognize occluded regions more effectively. In addition, we contend that the backbone network based on Darknet [26] and the feature fusion structure based on Path Aggregation Network (PAN) [27], commonly used in mainstream YOLO models, have relatively heavy parameter counts and computational demands. To achieve more effective model lightweighting, we propose a novel feature fusion network architecture (Section IV-B3).

III. DATASETS

A. A NOVEL MULTI-SCENE DATASET FOR BLIND INDIVIDUAL IDENTIFICATION

Currently, we have not yet found an open-source dataset specifically focusing on visually impaired individual features. So, we have expanded our previously created dataset [4] for blind individual identification to include a broader range of real-world scenarios. Table 1 shows the statistics for each target category in our dataset. The dataset now consists of 2,015 images, encompassing three recognizable categories: white canes for the visually impaired (including our simulated visually impaired individuals), long-handled umbrellas, and ordinary walking sticks. This dataset incorporates a significant number of images captured in low-light conditions, complex artificial lighting environments, and busy public intersections. Figure 1 presents examples of various new images from our new dataset.

Moreover, the dataset images include two widely-used types of white canes, over 150 distinct types of long-handled umbrellas, and ordinary walking sticks. Our dataset covers around 500 unique public environments, with variations in enough weather scenarios such as sunny, rainy, and snowy days. For each scene, images were taken from several perspectives: frontal, side, and rear angles, ensuring a comprehensive representation of the photographed person.

Compared to the dataset created in our previous study [4], as shown in Table 2, the new dataset has almost doubled the quantity for each object category and shooting scenario. Notably, we placed a strong emphasis on expanding the

FIGURE 1. Some example images from our new dataset showcase more challenging scenes (including low-light environments, complex artificial lighting conditions, and crowded public places).

collection of nighttime images, particularly those captured in low-light and complex lighting conditions. The previous dataset contained 175 images taken in low-light environments, whereas the new dataset includes 655 such images.

In addition to its scale and diversity, the proposed dataset poses several inherent challenges for object detection systems. First, a significant portion of the images are captured under low-light and complex artificial illumination conditions, where color distortion, reduced contrast, and noise severely degrade visual cues, particularly for thin objects such as white canes. Second, in crowded public environments, the target objects are frequently partially occluded by pedestrians, vehicles, or urban infrastructure, making it difficult for detectors to capture complete shape and contextual information. Third, visually similar objects, such as ordinary walking sticks and long-handled umbrellas, introduce strong inter-class interference, increasing the risk of false positives. Finally, variations in camera viewpoints and target scales further complicate detection, especially when the white cane occupies only a small portion of the image. These factors collectively make the dataset more challenging than conventional pedestrian or object detection benchmarks and better reflect real-world deployment scenarios.

This newly constructed dataset is characterized by its multi-scene, multi-region, and challenging condition diversity. To ensure transparency and facilitate future research, a comprehensive dataset card detailing the collection process, geographic distribution, device information, annotation protocol, and explicit data splits is provided in Section VIII.

Regarding the privacy protection measures in the dataset, all visually impaired individuals using white canes (or other handheld items) and other participants captured in the dataset have provided their consent to participate in this study. To further ensure privacy, the faces of bystanders appearing in public scenes were blurred. A comprehensive discussion of ethical considerations, potential mitigation for deployment, and privacy safeguards is provided in Section IX.

B. THE EXCLUSIVE DARK DATASET

The Exclusively Dark (ExDARK) dataset [12] is specifically designed for object detection and image enhancement in low-light environments, with its reliability validated by numerous object detection experiments. It comprises the largest collection of low-light images available to date, totaling 7,363 images captured under 10 distinct lighting conditions, ranging from dusk scenarios in the late afternoon to extremely dark settings at dawn. The dataset consists of annotations for 12 object categories, such as pedestrians, pets, cars, and bicycles, providing both image-level classification labels and localized object bounding boxes. We utilized this dataset to conduct a series of experiments aimed at evaluating whether our proposed model improvements exhibit superior recognition capabilities in general low-light object detection tasks.

IV. NETWORK ARCHITECTURE

A. EXISTING ISSUES AND IMPROVEMENT GOAL

According to our previous research [4], we identified that the current models exhibit poor recognition accuracy in low-light conditions, environments with limited visibility, and scenarios where the white cane is partially occluded. These challenges are common in outdoor object detection tasks. Furthermore, the pursuit of high detection accuracy has led to the adoption of models with maximum scaling factors, resulting in increased model complexity and higher computational demand. Model lightweighting has also become a critical requirement.

Therefore, our goal is to improve the model's recognition accuracy in low-visibility environments and occluded scenarios while making models at the same scale more lightweight.

Additionally, we aim to ensure that the proposed improvements are compatible with all mainstream convolutional neural networks.

B. OPTIMIZATION ARCHITECTURE

Our proposed end-to-end system, illustrated in Figure 2, consists of several optimized components that work collaboratively to address the challenges of blind individual identification in diverse outdoor scenarios. We have developed a comprehensive improvement framework for this object detection system by integrating a data augmentation network (Retinexformer Enhance), two distinct group convolution modules (CGblock and SEAM), and a CNN-based Cross-scale Feature Fusion (CCFF)-like feature fusion network. Figure 3 presents the schematic structure of the general object detection baseline framework with our used improvement integrated. This framework leverages the strengths of each component to enhance the overall detection performance.

To clarify the scope and rationale of the proposed optimization, the improvements are intentionally distributed across three key stages of the detection pipeline: the input stage, the backbone feature extraction stage, and the feature fusion stage.

At the input stage, a low-light enhancement network is introduced to mitigate illumination degradation and standardize image brightness in nighttime and complex lighting environments. At the backbone stage, grouped-convolution-based modules are employed to enhance feature extraction while reducing computational redundancy. Finally, at the feature fusion stage, a cross-scale feature fusion structure inspired by CCFF is adopted to improve multi-scale representation, while a lightweight SEAM attention module is further integrated to strengthen the modeling of small and partially occluded targets, and to enhance feature interaction across detection heads with minimal additional overhead.

These optimizations are jointly designed to address the key challenges of low-light conditions, occlusion, and model efficiency, while maintaining compatibility with mainstream one-stage object detection frameworks. In this section, we will introduce all the optimization measures we implemented in the input network, backbone network, and feature fusion stages.

1) INPUT-STAGE LOW-LIGHT ENHANCEMENT NETWORK

Data augmentation is effective for many specific object detection tasks. Traditional input data augmentation methods used in object detection systems, such as image flipping, scaling, Cut-mix, and Mosaic augmentation [18], generally show limited enhancement performance in specialized scenarios like low-light conditions.

To improve recognition accuracy in dark environments, we planned to incorporate a low-light image enhancement network as the input layer to standardize image brightness. We selected state-of-the-art image enhancement networks [14], [15], [16], [17] in recent years and compared their performance. We considered various factors, such as

TABLE 3. Comparative experiments on multiple low-light image enhancement modules were conducted based on the YOLOv8 system and the ExDARK dataset.

Method	Network	Params (M)	FLOPs (G)	$AP^{val}50$
Baseline		68.16	258.1	74.6%
+SCInet		68.23	258.8	75.1%
+Retiformer	YOLOv8x	68.30	260.3	75.3%
+IAT		68.25	265.6	75.0%
+GDIP		203.47	490.1	75.6%

network architecture, enhancement effectiveness, and computational complexity. As shown in Table 3, we experimented with various low-light image enhancement networks as input modules integrated into the object recognition system, evaluating their performance based on multiple parameters. Ultimately, we selected **Retinexformer** network [16] as the enhancement network for integration into different detection systems.

We evaluated multiple models across various parameters (as shown in the results in Table 3), the Retinexformer network emerged as the optimal choice overall. The integration of the Retinexformer network resulted in a 0.6% improvement in $AP^{val}50$ while adding less than 0.14M parameters to the entire system. Under comparable parameter and computational complexity, both the SCInet and IAT modules yield less improvement in accuracy for the YOLOv8 model compared to the Retinexformer network. In contrast, although the YOLOv8 model integrated with the GDIP module achieves higher recognition accuracy, it incurs significantly greater parameter counts and computational costs than the other network architectures.

We argue that the key advantage of Retinexformer network also lies in its architecture, which combines the classic Retinex-based algorithm [28] with self-attention modules and multi-head self-attention mechanisms [29]. Compared to traditional low-light enhancement networks based on the Retinex theory, it offers better suppression of image noise and artifacts. In contrast to enhancement systems that rely solely on convolutional neural networks, it achieves lower computational complexity while delivering superior image restoration quality.

2) GROUPED CONVOLUTIONS IN THE BACKBONE AND FEATURE FUSION

In addition to incorporating an image enhancement network at the input stage, introducing variant convolutions, such as grouped convolutions [30], is an effective approach to improving recognition accuracy in low-light environments and scenarios with partial occlusion. Grouped convolutions, such as the **Channelwise Convolution** [31] and **Depthwise Separable Convolution** [32] used in our optimization scheme, enhance the correlation between multiple sets of feature maps. This characteristic facilitates the fusion of local features with surrounding and global features in low-light scenarios, thereby improving recognition accuracy. In scenarios involving partial occlusion, this characteristic

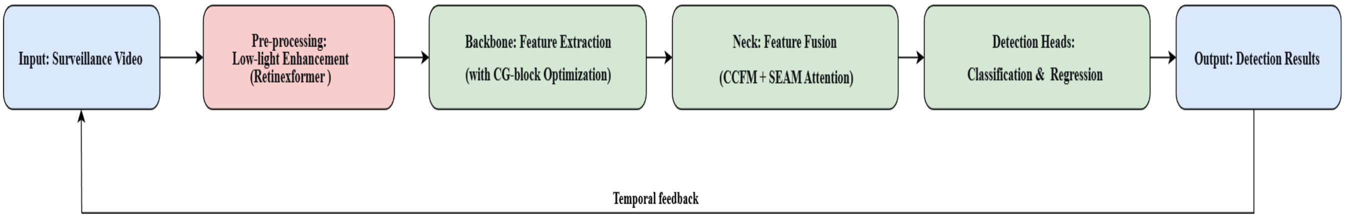


FIGURE 2. Overview of our proposed end-to-end framework for blind individual identification.

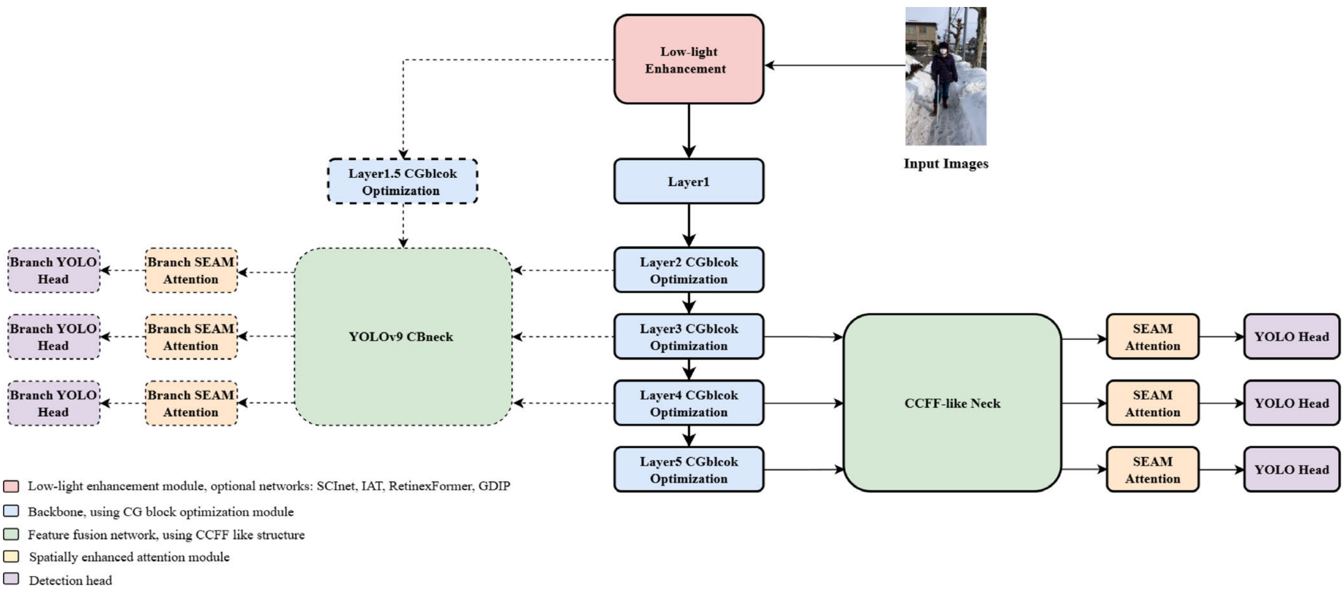


FIGURE 3. Overall optimization framework. The dashed-line components represent additional improvements specific to YOLOv9, while the solid-line components stand for the optimization structures applied to other baseline systems.

enables the model to learn the features of both the occluded and non-occluded regions, as well as their relationships, leading to improved occlusion detection performance.

Beyond improving recognition accuracy, these variant convolutions require less computational cost compared to conventional convolutional modules. Eqs. (1)-(3) respectively present the computational complexities of standard convolution, Channelwise convolution, and depthwise separable convolution.

$$C_{\text{std}} = H \cdot W \cdot K^2 \cdot C_{\text{in}} \cdot C_{\text{out}} \tag{1}$$

$$C_{\text{channelwise}} = H \cdot W \cdot K^2 \cdot C_{\text{in}} \tag{2}$$

$$C_{\text{DSC}} = H \cdot W \cdot (K^2 \cdot C_{\text{in}} + C_{\text{in}} \cdot C_{\text{out}}) \tag{3}$$

where H and W are the height and width of the input feature map, K is the kernel size (assumed square, $K \times K$), C_{in} and C_{out} are the numbers of input and output channels, and C is the computational cost measured in FLOPs.

Compared to Depthwise Separable convolution and standard convolution, Channelwise convolution offers low computational cost and high efficiency, making it an ideal replacement for the numerous basic convolution operations in complex backbone networks. Therefore, in this study, we replaced the standard convolutions within the CSPlayer (C2f) modules of the backbone network with Channelwise convolutions.

Depthwise Separable convolution, on the other hand, is essentially a composite module built by combining Channelwise convolution (Depthwise convolution) with Pointwise convolution. Although its computational complexity and parameter count are slightly higher than those of Channelwise convolution, it possesses the capability to extract both spatial and inter-channel features. This capability makes it more suitable for enhancing cross-layer and cross-channel feature connections during the feature fusion stage. However, due to the limited representational power of a single Depthwise Separable convolution and its relatively simple fusion ability, it may not adequately handle complex fusion requirements. Following previous work [21], we employed a multi-scale module based on Depthwise Separable convolution for application in the feature fusion network in this study.

a: CHANNELWISE CONVOLUTION-BASED C2f MODULE

Although Channelwise convolution has a low computational cost, simply replacing the standard convolutions in C2f can compromise the backbone's feature extraction capability. We drew inspiration from the **Context Guided (CG) Block** in CGNet [20], replacing the bottleneck in the C2f module with a CG block.

The operations of the CG block are formally expressed in Eqs. (4)-(6). The CGblock employs standard Channelwise convolution to extract local features, while utilizing Channelwise Dilated convolution to capture surrounding contextual features, followed by feature fusion.

$$X_{\text{local}} = \text{Channel-wise Conv}_{3\times 3}(X) \tag{4}$$
$$X_{\text{surround}} = \text{Channel-wise DConv}^{r}_{3\times 3}(X) \tag{5}$$
$$X_{\text{fused}} = \text{BN}(\text{PReLU}(X_{\text{local}} + X_{\text{surround}})) \tag{6}$$

where X is the input feature map, Channel-wise Conv and Channel-wise DConv are the Channelwise convolution and Channelwise Dilated convolution, and r is the dilation rate.

Compared to traditional bottlenecks composed of multiple layers of standard convolutions, the CGblock is more parameter-efficient and offers superior feature extraction capabilities. As shown in Fig. 3, the optimized module replaces the C2f modules in stages 2, 3, 4, and 5 of the backbone network, as well as one stage of the branched backbone in YOLOv9, indicated by the dashed lines.

Unlike other detection systems we use, the C2f module in YOLOv9 [33] employs a unique **RepNBottleneck**, which incorporates a lightweight convolution layer called RepConv [34]. To preserve YOLOv9's original innovations while incorporating our optimized structure, we designed a hybrid module called **RepNCGBlockBottleneck**. This module integrates the global information extraction component featuring Channelwise convolution with the RepConv convolutional layer. Figure 4 shows more details of the structure. We incorporate the fundamental module of the CG block (Channelwise Conv + Channelwise Dilated Conv + BN + ReLU + FC) as a parallel branch alongside the RepConv branch in YOLOv9 C2f. This enhances the module's ability to extract both local and global features. The hybrid module is fully compatible with all object detection systems employing the RepNBottleneck structure, ensuring seamless adaptation.

b: DEPTHWISE SEPARABLE CONVOLUTION IN FEATURE FUSION

We adopted the **Spatially Enhanced Attention Module (SEAM)** from the YOLO-Face system [21]. SEAM utilizes three Depthwise Separable convolution modules of varying sizes to enhance feature representation. Due to the high computational demand of the full multi-head SEAM (three branches), our optimized framework adopts only a single branch as the attention mechanism. The single-branch SEAM structure is illustrated in Figure 5.

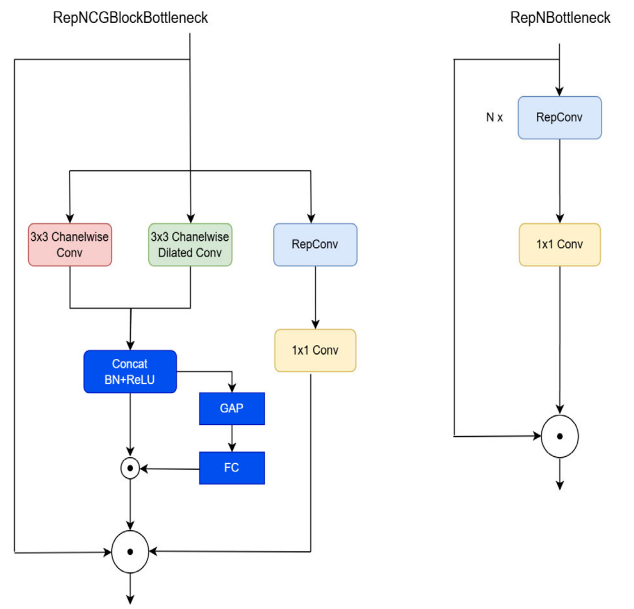


FIGURE 4. Compared to the standard RepNBottleneck structure, we designed the simplified CGblock to run in parallel with the RepConv layer, and then combined the feature layers obtained from both.

As shown in Figure 3, we integrate this attention mechanism at three output locations during the feature fusion stage and between the three detection heads. Features at different scales from each detection head are captured by the attention mechanism and processed through a depthwise separable convolution module, which learns the relationships between occluded and unoccluded regions, thereby enhancing the model's ability to learn from occluded features. For YOLOv9 [33], which features a branched backbone network, we applied this attention mechanism to all six detection heads.

3) CNN-BASED CROSS-SCALE FEATURE FUSION STRUCTURE

Most state-of-the-art (SOTA) object detection systems have continued to adopt the Feature Pyramid Network (FPN) and Path Aggregation Network (PAN) [27] frameworks for feature fusion. This combination structure delivers commendable high performance and efficient convergence speed but imposes constraints on the computational cost of the feature fusion component. The RT-DETR system [35] introduced a novel feature fusion module, the **CNN-based Cross-scale Feature Fusion** (CCFF), which redefines feature fusion paths to better integrate fine-grained details and contextual features, significantly enhancing overall model performance. Additionally, we found, unlike the conventional FPN+PAN feature fusion pyramid architecture [27], the Cross-scale Feature Fusion structure incorporates convolutional layers during feature output to enhance dimensionality and reduce channel count. In traditional feature fusion pyramids, the number of channels at each input level is determined by the corresponding feature extraction layers in the backbone

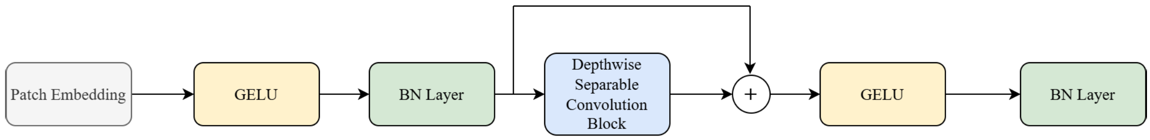


FIGURE 5. The structural diagram of the SEAM in this study.

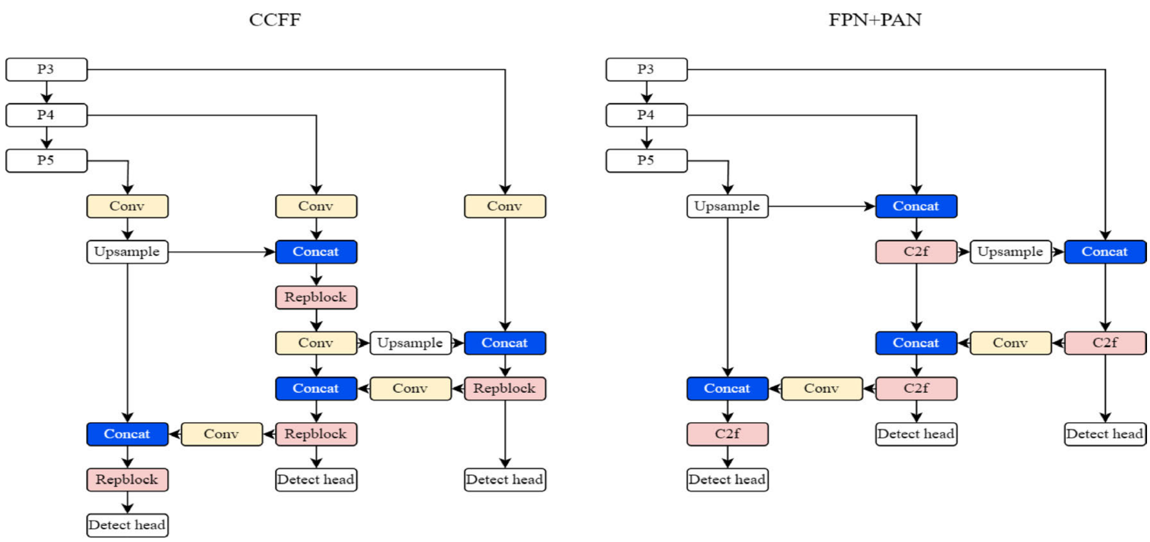


FIGURE 6. The CCFF structure in RT-DETR and the FPN+PAN structure in commonly object detection systems (YOLOv8).

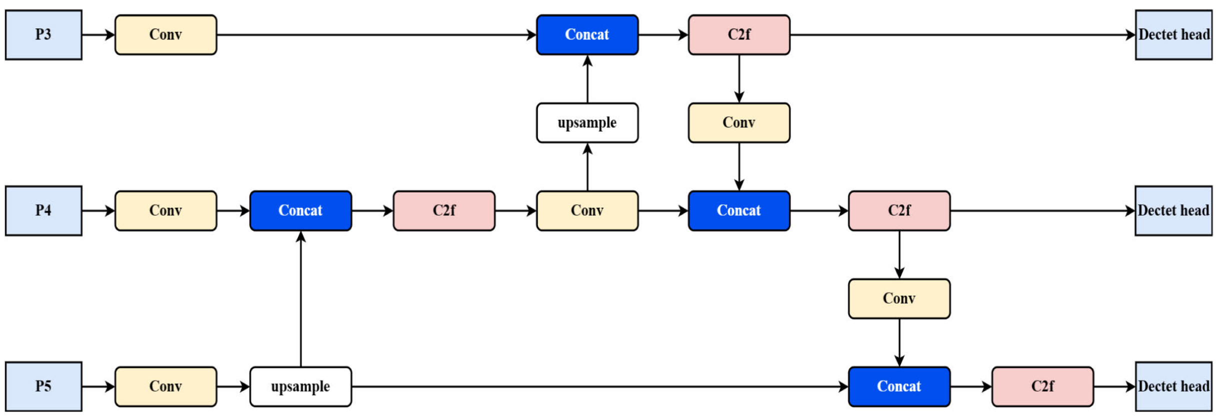


FIGURE 7. Illustration of the CCFF-like framework. The various basic modules in this diagram only represent basic units with similar functions.

network, typically increasing in the order of 128, 256, 512, and 1024. In contrast, the CCFF network introduces convolutional layers at the input stage to standardize the number of channels across all input features (fixed to 256 in this study), significantly reducing computational overhead. The structural differences between the two are illustrated in Figure 6. In the subsequent experimental section, we confirmed that using this structure effectively reduces the model's parameter count and computational complexity, without negatively impacting recognition accuracy. Additionally, the fusion of multiple scales proves more effective for detecting small targets and handling partially occluded objects.

We learned CCFF principles while maintaining YOLO detection system's original concat, upsampling, and convolutional modules. We designed CCFF-like feature fusion structures tailored for each baseline system we used in the experiments to optimize feature integration. Figure 7 shows an abstract framework of the CCFF-like structure that we adapted to various systems.

V. EXPERIMENTS

In this section, we conduct evaluation experiments using our created dataset and the ExDARK dataset. First, we outline the processing details of the two datasets before training. Then we present the training details for multiple comparative experiments and ablation studies.

A. DATASET PROCESSING

The dataset we created, along with the ExDARK dataset, utilizes an annotation approach akin to the PASCAL VOC object annotation methodology [36]. Prior to utilization, all annotation files are converted into a format that is compatible with object detection system labeling requirements.

Within our dataset, we divided the three types of annotated images into training and validation groups respectively in a 9:1 ratio. For the ExDARK dataset, experiments were conducted in adherence to its pre-established training and validation set partitions.

B. IMPLEMENTATION DETAILS

This study leverages several state-of-the-art (SOTA) object detection systems as baselines, namely YOLOv8 [37], YOLOv9 [33], YOLOv10 [38], YOLOv11 [39], YOLOv12 [40] and RT-DETR [35]. YOLOv8, the earliest among these, was released in January 2023, while YOLOv12, the latest, debuted in February 2025. All models (including our models) were developed based on these detection frameworks.

Experiments were conducted on a machine equipped with an NVIDIA RTX 4090 GPU, using PyTorch as the development framework. To achieve optimal accuracy, since recognition speed is not a priority for this study, all models were trained with the maximum scaling factor (YOLOv9 utilizes YOLOv9c, while the other models adopt the x scaling factor). Uniform hyperparameters were applied across all systems, including a batch size of 8, 300 epochs, and SGD [41] as the gradient optimizer.

Mean Average Precision (mAP) was adopted as the primary performance metric to compare the improved models with the native versions of these SOTA systems. Training and validation were initially performed on our created dataset. Subsequently, the same training and validation settings were applied to the ExDARK dataset to demonstrate the robustness of the proposed improvements in outdoor challenging scenarios.

To assess the statistical reliability of our results while addressing computational constraints, we implemented a test set partitioning strategy exclusively on the ExDARK dataset. The partitioning procedure was conducted as follows:

1. Partition Methodology: The original ExDARK test set was randomly divided into three mutually exclusive subsets of approximately equal size, while maintaining the original class distribution across each partition through stratified sampling.

TABLE 4. Comparison of various object detectors performances based on our self-create dataset (The models with the "OP" suffix in the table represent our optimized versions).

Model	Params (M)	FLOPs (G)	FPS	Category	$AP^{val}50$	$AP^{val}50\text{-}95$
YOLOv8x	68.2	258.1	510	average	94.1%	60.5%
				blind	95.3%	60.1%
				stick	97.3%	61.1%
				umbrella	91.0%	60.5%
YOLOv8x-OP	**47.4**	**217**	229	average	**95.6%**	**61.8%**
				blind	**98.5%**	59.4%
				stick	96.0%	59.6%
				umbrella	**92.2%**	**66.5%**
YOLOv9c	51	238.9	276	average	93.7%	61.3%
				blind	98.9%	60.6%
				stick	95.3%	60.4%
				umbrella	86.7%	62.8%
YOLOv9c-OP	**43.4**	**222.6**	157	average	**94.9%**	59.2%
				blind	97.4%	57.9%
				stick	**96.5%**	58.5%
				umbrella	**90.8%**	61.2%
YOLOv10x	31.7	171	397	average	89.3%	56.9%
				blind	95.1%	55.5%
				stick	92.6%	61.2%
				umbrella	80.0%	54.0%
YOLOv10x-OP	**26**	**165.3**	244	average	**91.9%**	**56.9%**
				blind	**96.3%**	55.4%
				stick	91.5%	53.8%
				umbrella	**87.7%**	**61.5%**
YOLOv11x	56.9	195.5	349	average	93.8%	60.7%
				blind	97.5%	59.6%
				stick	98.1%	64.7%
				umbrella	85.6%	57.7%
YOLOv11x-OP	**36.6**	**138.1**	236	average	**95.2%**	60.2%
				blind	**98.1%**	59.5%
				stick	96.5%	58.9%
				umbrella	**90.8%**	**62.2%**
YOLOv12x	59.1	199.8	347	average	91.8%	57.9%
				blind	98.0%	58.6%
				stick	91.2%	55.3%
				umbrella	86.2%	59.8%
YOLOv12x-OP	**43.3**	**147.6**	219	average	**92.7%**	**58.5%**
				blind	**98.1%**	58.5%
				stick	**93.8%**	**56.3%**
				umbrella	**86.3%**	**60.6%**
RT-DETR-R101	76	259	127	average	87.1%	55.5%
				blind	94.4%	54.8%
				stick	89.8%	58.4%
				umbrella	77.1%	53.4%

2. Quality Assurance: Each partition contained representative samples from all 12 object categories and 10 lighting conditions present in ExDARK.

3. Evaluation Protocol: Our pre-trained models (trained on the standard ExDARK training set) were evaluated separately on each of the three test partitions, with each partition tested three times using random image orders to assess performance consistency. The performance metrics from these nine evaluations (three partitions $\times$ three repetitions) were used to compute the mean and standard deviation reported in our results.

Limitations of the Evaluation Approach: While our enhanced evaluation strategy with nine measurements provided more robust statistical estimates than a single test run, we acknowledged that this approach still had inherent limitations. The primary constraint was that all evaluations were performed on the same pre-trained model, thus the reported variance primarily reflected performance variation across different test subsets rather than capturing the full model training instability that would have been observed in multiple independent training runs. Nevertheless, this method offered a practical compromise under computational constraints and provided valuable insights into model consistency across diverse data samples.

This partitioning approach was applied only to the ExDARK dataset due to its sufficient size (7,363 images total), which allows for meaningful statistical analysis when divided. Our created dataset (2,015 images) was deemed too small for reliable partitioning, as creating smaller subsets would risk significant information loss and unstable variance estimates.

To isolate the effects of accuracy improvement and model lightweighting, ablation experiments were conducted on the ExDARK dataset, testing individual components of the proposed architecture.

VI. RESULT ANALYSIS

A. COMPARISON ON OUR BLIND INDIVIDUAL FEATURE DATASET

1) QUANTITATIVE COMPARISON

Table 4 illustrated the recognition performance of various object detection models trained and tested on our created dataset. As shown, even without applying our optimization methods, mainstream object detection systems can already achieve high-accuracy recognition when trained on our dataset. In comparison, our optimized models demonstrate a significant reduction in both parameter count and computational requirement. For example, the most substantial savings were observed in YOLOv8x and YOLOv11x, with parameter counts reduced by 30.5% (20.8M parameters) and 35.7% (20.3M parameters) respectively. Even the most modest reduction, seen in YOLOv9c, still represented a 14.9% (7.6M parameters) decrease. Moreover, in terms of model accuracy, our optimized models exhibited consistent improvements in $AP^{val}50$, ranging from 1.0 to 2.6 percentage points across different architectures. For instance, YOLOv10x showed the most substantial gain, improving from 89.3% to 91.9% (+2.6 percentage points), while YOLOv8x increased from 94.1% to 95.6% (+1.5 percentage points). This indicates that our optimization approach not only lightens detection models significantly but also enhances their detection accuracy.

While the integration of enhancement and attention modules introduced additional computational overhead, resulting in approximately 40% reduction in inference speed compared to the baseline models, the absolute performance remained more than sufficient for practical deployment. As indicated by the FPS metrics in Table 4, our optimized models achieved inference speeds well above 100 FPS. We asserted that this level of performance does not impair the real-time detection of blind individuals in public surveillance scenarios. The temporal frequency at which a visually impaired person enters and moves through a camera's field of view is relatively low. Therefore, a processing speed of 100 FPS provided a substantial margin, ensuring that the system can not only track the individual's movement smoothly but also perform secondary analysis or maintain a buffer for processing peak loads. The primary goal in this context was not maximizing frame rate, but achieving a sufficient frame rate coupled with high detection reliability, a balance which our method successfully delivers.

(A) (B) (C) (D)

FIGURE 8. Comparison of baseline model and optimization model in challenging scenarios (A, C are the baseline model recognition results of YOLOv8x, B, D are the optimization model recognition results).

Furthermore, we incorporated the DETR-style RT-DETR-R101 model into our experiments. The results, presented in Table 4, revealed a notable finding: the RT-DETR-R101 model, despite its significantly larger parameter count and computational complexity, did not achieve superior accuracy compared to the more recent YOLO variants on our experimental datasets. A plausible explanation for this was the relatively small scale of our training data. Transformer-based models typically require vast amounts of data to unlock their full potential and avoid overfitting. In data-constrained environments like ours, the inherently more parameter-efficient and localization-optimized convolutional architectures of modern YOLO variants demonstrate a clear advantage.

2) DETAILED ANALYSIS AND VISUAL ASSESSMENT

To provide comprehensive insights into the performance characteristics of our proposed framework, we conducted a detailed visual analysis across various challenging scenarios.

1. Performance Advantages of Optimized Models

Figure 8 presented side-by-side comparisons between baseline and optimized models on our created dataset. The visual evidence consistently showed that our optimized framework delivers more reliable detection across diverse conditions.

Enhanced Low-light Performance: In twilight conditions, the baseline YOLOv8x model frequently produced false negatives or low-confidence detections (Figure 8 A), whereas our optimized model successfully identified the white cane with high confidence (Figure 8 B).

Improved Occlusion Handling: For partially occluded targets, the baseline model often produced low-confidence detection, while our model, equipped with the SEAM attention mechanism, demonstrated robust performance despite partial visibility (Figure 8 C-D).

2. Scenarios with Reliable Performance

Analysis of the additional scenes shown in Figure 9 indicated that the optimized framework achieves robust performance in several key scenarios:

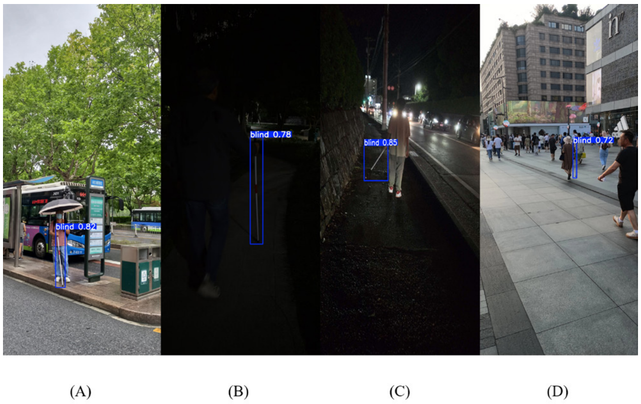


FIGURE 9. Performance of YOLOv8x optimization model in more challenging scenarios.

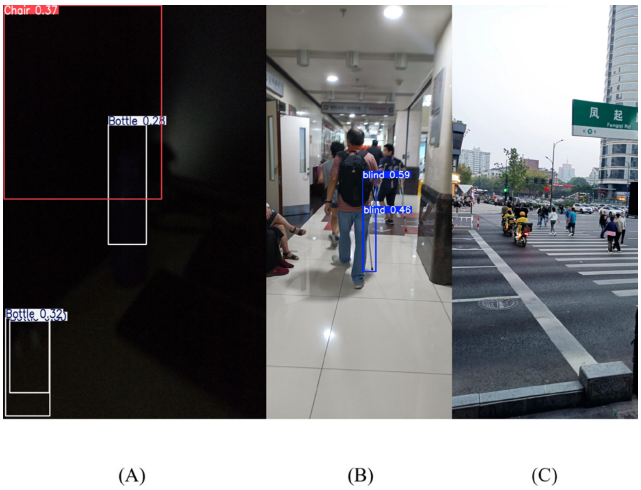


FIGURE 10. The recognition errors and non recognition of our optimization model in extreme scenarios (incorrect recognition of the chair and bottle in the upper left corner in Figure A, incorrect recognition of most of the occluded white cane in Figure B, and non recognition due to the target size being too small in Figure C).

Daytime Conditions: Under sufficient lighting conditions, our optimized model maintained high accuracy in recognition and has good anti-interference effect on similar objects (Figure 9 A).

Moderate Low-light Environments: In scenarios with reduced but not extreme lighting (e.g., dusk, shaded areas), our model still maintained high recognition accuracy (Figure 9 B-C).

Partial Occlusion: When the white cane was partially obscured (approximately 15-50% occlusion), our model maintained detection capability, leveraging the enhanced feature representation learned through our architectural improvements (Figure 9 D).

3. Remaining Challenges and Failure Cases

Despite the overall improvements, we summarized the current situation where the model still cannot accurately identify based on the results of the Exdrak dataset and our created dataset:

Extreme Darkness: In near-total darkness where the target object occupies very few pixels or lacks discernible features, neither the enhancement network nor the detection backbone could reliably identify the target (Figure 10 A).

Severe Occlusion: When the cane was heavily obscured (>70% occlusion), the available visual cues became insufficient for confident detection, even with our attention mechanisms (Figure 10 B).

Very Small Targets: At large distances where the target object represents less than 0.1% of the image area, the current multi-scale feature representation failed to capture the necessary discriminative features (Figure 10 C).

B. COMPARISON ON THE EXDARK DATASET

1) LARGE-SCALE MODEL EVALUATION

Table 5 presented the test results of various detection models trained on the ExDARK dataset. Similar to the results of our previous experiment, the optimized models demonstrated clear advantages over their original versions. For instance, the optimized YOLOv10 model trained on ExDARK achieved a 4.1% improvement in $AP^{val}50$ compared to the original YOLOv10x model, while reducing the parameter count by 18%. The consistently low standard deviation values (ranging from 0.1% to 0.4% for $AP^{val}50$ and $AP^{val}50$-95 across different models) indicated that the observed accuracy gains were robust and not heavily dependent on specific subsets of the test data. In addition, similar to the test effect on the dataset created by us, the RT-DETR model was also limited by the size of the dataset, and the detection accuracy was lower than the latest Yolo series models and our optimized models. These findings confirmed that our optimization approach is robust even in general low-light outdoor scenarios.

Compared to most traditional one-stage object detection systems, YOLOv9 performed poorly in terms of model lightweighting due to its branching backbone architecture. Our lightweight strategy primarily focused on improving the feature enhancement network and the fundamental bottleneck modules, whereas YOLOv9c was characterized by its use of two feature extraction networks. Additionally, during our experiments, we found that the YOLOv9e model (the model with the largest scaling level), featuring a complete dual backbone and dual feature enhancement networks, was not suitable for incorporating additional structures. An excessively deep layer structure easily led to the vanishing of gradients during training. Therefore, we ultimately selected the YOLOv9c model (the second-largest scaling level) with a relatively smaller branch backbone scale as the baseline for all experiments.

2) LIGHTWEIGHT MODEL EVALUATION

In addition to evaluating the performance of larger models, we also considered potential use cases for lightweight models. Therefore, we conducted comparative experiments using lightweight (s-size) models as well. The training hyperparameters for the lightweight models were kept consistent with those used for the larger models. Table 6 presented the evaluation results on the Exclusive Dark dataset (The lightweight version of YOLOv9 did not utilize components such as the branched backbone or branched detection heads found in the larger model. RT-DETR had not released an ultra-lightweight model. Therefore, they were not included in this experiment). Our optimized structure brought even greater accuracy improvements to lightweight models compared to larger ones, with an average AP increase of 2.5% across most detection models. Moreover, we achieved further model compression; for instance, in models such as YOLOv8s, our optimized structure reduced the number of parameters by up to 55% (from 11.2M to 6.6M) while enhancing performance. Naturally, given the inherently low computational demands of lightweight models, the additional overhead introduced by the Retinexformer network and SEAM structure could not be overlooked. On average, the optimized lightweight models incurred approximately 3 GFLOPs more than their baseline counterparts. However, considering the substantial improvements in accuracy and the significant reduction in parameter count brought by our optimized structure, we believed this level of increase in computational cost is acceptable. The results indicated that our proposed optimization framework can enhance the baseline model in most aspects, even when applied to lightweight models.

A noteworthy observation was the relatively more pronounced impact of our optimization structures on inference speed compared to the large-scale models. This could be attributed to the fact that components like the Retinexformer enhancement network and SEAM attention module had fixed computational costs that do not scale with model size. Consequently, when integrated into already compact architectures, these fixed overheads represented a larger proportional burden on the overall computational budget. However, despite this amplified relative impact, all optimized lightweight models still achieved inference speeds exceeding 300 FPS on desktop-grade hardware. This performance far surpassed the requirements for real-time detection in public surveillance scenarios. Therefore, while the percentage reduction in speed is more noticeable for lightweight models, the absolute performance remained more than adequate for practical deployment, ensuring that detection efficacy was uncompromised while still benefiting from the accuracy improvements demonstrated by our optimization framework.

The current experiments were conducted on a high-performance NVIDIA GeForce RTX 4090 GPU. In the future, we will incorporate edge devices such as GPUs from the NVIDIA Jetson series for testing.

C. ABLATION STUDIES

In this section, we conducted individual tests for each optimization structure within our optimization framework to evaluate its specific contribution to the overall system improvement. We added these structures separately to the YOLOv8 system and evaluated them on the Exclusive Dark dataset. The results were summarized in Table 7.

TABLE 5. The performance of various system models on the Exclusive Dark dataset.

Model	Params (M)	FLOPs (G)	$AP^{val}50$	$AP^{val}50\text{-}95$	FPS
YOLOv8x	68.2	258	74.6±0.2%	48.4±0.1%	545
YOLOv8x-OP	**47.5**	**217**	**75.4±0.1%**	**49.3±0.1%**	250
YOLOv9c	51	239	78.9±0.3%	51.8±0.3%	348
YOLOv9c-OP	**43.7**	**226**	**79.1±0.3%**	**52.7±0.2%**	199
YOLOv10x	31.7	171	68.8±0.3%	43.9±0.2%	517
YOLOv10x-OP	**26.1**	**161.4**	**72.9±0.2%**	**47.2±0.2%**	266
YOLOv11x	56.9	195.5	74.7±0.2%	48.7±0.1%	529
YOLOv11x-OP	**36.6**	**138.1**	**75.0±0.2%**	**48.8±0.1%**	306
YOLOv12x	59.1	199.9	75.0±0.3%	48.9±0.2%	429
YOLOv12x-OP	**34.6**	**147.6**	**75.2±0.3%**	**49.0±0.1%**	249
RT-DETR-R101	**76**	**259**	71.1±0.3%	45.1±0.3%	127

Note: Performance metrics are reported as mean ± standard deviation across three distinct partitions of the ExDARK test set.

TABLE 6. The performance of the lightweight models on the Exclusive Dark dataset.

Model	Params (M)	FLOPs (G)	$AP^{val}50$	$AP^{val}50\text{-}95$	FPS
YOLOv8s	11.2	28.8	71.5±0.3%	41.9±0.2%	2100
YOLOv8s-OP	**6.6**	31.8	**74.3±0.2%**	**43.6±0.1%**	347
YOLOv10s	8.1	24.8	69.5±0.4%	44.0±0.2%	1718
YOLOv10s-OP	**5.2**	27.6	**71.9±0.3%**	**44.3±0.2%**	320
YOLOv11s	9.5	21.7	71.1±0.2%	45.2±0.2%	1666
YOLOv11s-OP	**6.4**	25.4	**72.6±0.2%**	45.1±0.1%	318
YOLOv12s	9.2	21.5	69.0±0.3%	43.2±0.3%	1736
YOLOv12s-OP	**5.1**	24.9	**70.1±0.2%**	**44.0±0.2%**	319

Note: Performance metrics are reported as mean ± standard deviation across three distinct partitions of the ExDARK test set.

TABLE 7. The respective effects of each basic optimization module on the YOLOv8x model and the Exclusive Dark dataset.

Method	Network	Params (M)	FLOPs (G)	FPS	$AP^{val}50$	$AP^{val}50\text{-}95$
Baseline	YOLOv8x	68.2	258	545	74.6±0.2%	48.4±0.1%
CGblock		67.8	251	**562**	74.9±0.2%	48.6±0.1%
+Retiformer		68.3	260.3	259	**75.3±0.3%**	**49.1±0.2%**
+SEAM		68.4	259.9	344	74.8±0.1%	48.5±0.1%
CCFF		**47.3**	**215**	257	74.8±0.2%	48.6±0.1%
Baseline	YOLOv9c	51	239	348	78.9±0.3%	51.8±0.3%
CGblock		**50.4**	**236.2**	**355**	**79.0±0.3%**	**51.9±0.2%**

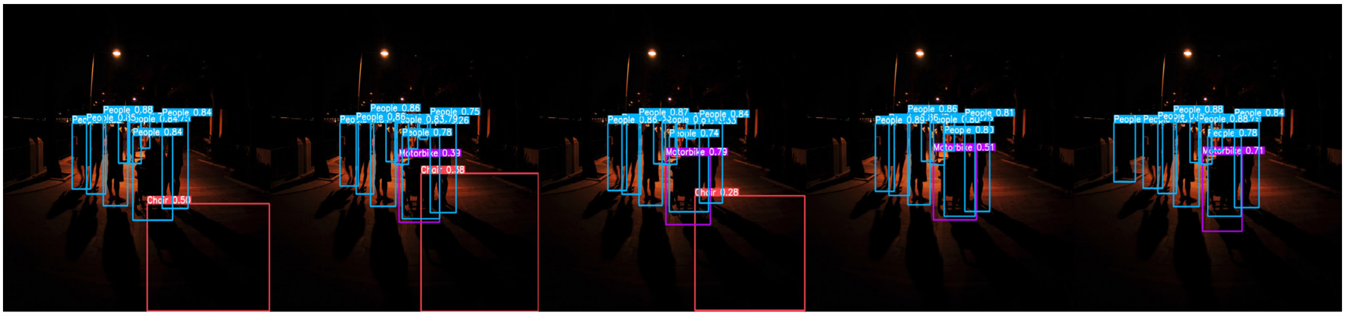
The feature fusion pyramid structure designed with a CCFF-like approach proved to be the most effective in reducing parameter count and computational demand. Its standalone design resulted in a 30.6% reduction in parameter count (from 68.2M to 47.3M) and a 16.7% decrease in computational demand (from 258G FLOPs to 215G FLOPs) when applied to the YOLOv8 baseline model. Channelwise convolution layers, compared to standard convolution layers, achieved approximately a 1% reduction in parameter count while also improving $AP^{val}50$ from 74.6% to 74.9%. Additionally, the standalone integration of the Retinexformer data enhancement network resulted in a 0.7% increase in $AP^{val}50$, with the most notable improvements observed in low-light scenarios. The SEAM attention mechanism also contributed a 0.2% improvement in $AP^{val}50$, while introducing only 0.2M additional parameters and 1.9 GFLOPs of computational overhead to all systems.

Compared to the baseline YOLOv8x (545 FPS), the integration of different modules demonstrated varying effects on inference speed. The Retinexformer enhancement network introduced the most substantial computational overhead,

TABLE 8. Comprehensive step-wise ablation study on ExDARK dataset showing cumulative effects of all proposed components (based on YOLOv8x).

Model Configuration	Retinexformer	CG-block	SEAM	CCFM	Params (M)	FLOPs (G)	$AP^{val}50$	$AP^{val}50\text{-}95$	FPS
Baseline	×	×	×	×	68.2	258.1	74.6±0.2%	48.4±0.1%	545
+ Retinexformer	✓	×	×	×	68.3	260.3	75.3±0.3%	49.1±0.2%	259
+ CG-block	✓	✓	×	×	67.9	253.2	75.3±0.2%	49.2±0.2%	268
+ SEAM	✓	✓	✓	×	68.1	255.1	75.3±0.1%	49.3±0.1%	242
+ CCFM	✓	✓	✓	✓	**47.5**	**217.0**	**75.4±0.1%**	**49.3±0.1%**	250

Note: Symbols ✓ and × indicate the inclusion and exclusion of each component, respectively.



(A) (B) (C) (D) (E)

FIGURE 11. **Illustration of the results of ablation experiment. The models from left to right are: (A) Baseline (YOLOv8x), (B) Baseline+Retinormer, (C) Baseline+Retinormer+CG Block, (D) Baseline+Retinormer+CG Block+SEAM, and our proposed complete model ((E) Baseline+Retinormer+CG Block+SEAM+CCFM).**

reducing speed to 259 FPS due to its intensive image restoration operations. Similarly, the CCFF feature fusion structure also significantly impacted performance (257 FPS) as it redefines the entire feature integration pathway. In contrast, the SEAM attention mechanism showed relatively modest speed reduction (344 FPS), indicating its efficiency in providing occlusion awareness. Notably, replacing standard convolutions with CG blocks actually improved inference speed (562 FPS), demonstrating that our architectural optimization not only enhanced feature extraction but also achieved computational efficiency through grouped convolution designs.

As shown in Table 8, our proposed optimization modules provided significant cumulative performance gains. This demonstrated the synergistic effectiveness of our components in balancing accuracy improvements with model efficiency. Only through the combination of all components was achieved a 0.8% improvement in $AP^{val}50$ accuracy, while reducing the number of parameters by 30.3% and the computational load by 15.9%.

Figure 11 exemplifies the detection results presented in Table 8. As shown in the figure, the stacking of our optimization modules reduced erroneous recognition (object Chair in red box) in dark environments and improved the overall recognition accuracy (objects People and Motorbike in blue and purple boxes, respectively).

We also evaluated the performance of the proposed RepNCGBlockBottleneck module. As shown in the results presented in Table 7, compared to the original YOLOv9c model, the optimized model with the RepNCGBlockBottleneck module integrated into the backbone achieved a 0.1% improvement in $AP^{val}50$. Additionally, both the parameter count and computational requirements were reduced to a certain extent. These outcomes align with our optimization objectives.

VII. CONCLUSION

This study developed a more comprehensive multi-scene dataset tailored for blind individual recognition tasks. To address the challenges of low recognition accuracy in low-light and occlusion scenarios, we proposed an enhanced solution compatible with most current one-stage object detection systems. Our approach integrates the Retinexformer low-light enhancement network and various group convolution structures, while also optimizing the number of layers, channels, and specific paths within the feature fusion pyramid network.

We evaluated our dataset using several mainstream object detection systems from the past two years, all of which produced high-accuracy recognition models. When training these systems with our optimized solution, the results demonstrated significant improvements in model lightweighting and accuracy compared to the official baseline models. Additionally, experiments conducted on the ExDARK public dataset further confirmed the feasibility of our optimization approach, yielding similar performance benefits.

Future research will pursue several important directions to address the limitations identified in this study. First, we will focus on overcoming the current challenges in extreme scenarios, including near-total darkness environments, heavily occluded targets (>70% occlusion), and very small-scale objects. Second, we plan to developing audio-assisted measures for public spaces. Finally, we will create more comprehensive datasets that encompass diverse disability recognition tasks beyond visual impairment, enabling the development of more inclusive and universally accessible assistive technologies.

VIII. DATASET AND CODE AVAILABILITY

A. LICENSE AND RELEASE PLAN

To foster further research in assistive technologies for the visually impaired and ensure responsible use, the dataset and the source code for the proposed optimization framework will be made available to the academic community upon reasonable request.

- Availability: Interested researchers are encouraged to contact the corresponding author via email at [jihaotian2019@outlook.com] to request access. Requests will be reviewed to ensure they align with the intended academic and non-commercial use.
- Dataset License: The dataset will be provided under the Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0) license.
- Code License: The source code will be provided under the MIT License.

B. DATASET CARD

A comprehensive dataset card is provided below to ensure transparency and reproducibility. This card complements the statistical overview of the dataset provided in Section III.

- Collection Process and Geography: Images were collected across diverse public spaces in multiple geographic locations, including Hangzhou, China, and Kumamoto and Yamagata Prefectures, Japan. This multi-region collection strategy ensures that the dataset encompasses variations in urban infrastructure, architectural styles, and general environment, thereby enhancing the robustness and generalizability of models trained on it. The collection process simulated the natural walking patterns of visually impaired individuals and was conducted with strict adherence to ethical guidelines in both countries.
- Devices: Images were captured using a variety of consumer-grade devices to ensure diversity, primarily using smartphone cameras (including iPhone and Android models). This approach guarantees that the dataset is representative of real-world scenarios where assistive systems might operate.
- Time-of-Day Mix: A significant emphasis was placed on capturing a diverse time-of-day and lighting condition mix. As detailed in Section III, the dataset includes 655 low-light images captured during dusk, night, and dawn, alongside images from daytime and sunny conditions, covering all typical outdoor lighting variations.
- Annotation Protocol and Quality Assurance: The annotation process followed the PASCAL VOC format. All bounding boxes were meticulously drawn by human annotators to tightly enclose the target objects (blind stick, normal stick, umbrella).
- Explicit Data Splits: The dataset is explicitly partitioned into training and validation sets to facilitate fair comparison. The splits are performed at the image level in a 9:1 ratio (training: validation). Specifically, the 2,015 images are divided into approximately 1,814 images for training and 201 images for validation. This validation set also serves as the test set for final evaluation, a common approach for datasets of this scale [12], [36]. The splits are stratified to maintain a similar distribution of object categories, geographic locations, and challenging scenarios (e.g., low-light, occlusion) across both sets.

IX. ETHICAL CONSIDERATIONS AND PRIVACY SAFEGUARDS

Given the proposed application within public surveillance systems, we recognize the paramount importance of addressing ethical concerns and implementing robust privacy safeguards. This section outlines our commitments and recommendations for the responsible development and deployment of such assistive technology.

- Ethical Approval and Informed Consent: This study was conducted in accordance with the ethical standards of our institution. All individuals depicted in our self-constructed dataset, including visually impaired participants and bystanders in public scenes, provided written informed consent for their data to be collected and used for this research purpose. To further protect the privacy of non-consenting individuals who may appear in public background, all identifiable faces of bystanders have been systematically blurred in the dataset.
- Mitigation of Potential Harms from False Positives: We acknowledge that a false positive detection (e.g., mistakenly identifying an ordinary walking stick as a white cane) could lead to unnecessary alerts or a misallocation of assistive resources. To mitigate this risk:
  1.We have intentionally included "normal stick" and "umbrella" as interference categories during training to enhance the model's discriminative capability and reduce false alarms on common confounding objects.
  2.The confidence threshold for triggering an alert can be set high in a deployment setting to prioritize precision over recall, ensuring that only high-confidence detections are brought to an operator's attention.
- Recommended Safeguards for Deployment: For any future real-world implementation, we strongly recommend the following safeguards be put in place:
  1. On-Device/On-Server Processing: The detection algorithm should run on the local server of the

surveillance system, ensuring that video data does not leave the secure local network. No raw footage or personal data should be transmitted to external cloud services.

2. Data Retention Policies: Raw video footage should not be stored permanently. After the AI processing is complete and a relevant alert is resolved, the source data should be automatically deleted after a short, pre-defined period (e.g., 24-48 hours), unless required for auditing or security incidents unrelated to this system.

3. Transparency and Public Signage: Areas where this assistive technology is active should be clearly marked with public signage to inform citizens that computer vision technology is in use for the purpose of assisting visually impaired individuals. This promotes public awareness and trust.

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